

Strategy 2: Suppose you want to solve

$$\int \tan^m(x) \sec^n(x) dx.$$

1. If n is even, save a factor of $\sec^2(x)$ and use $\sec^2(x) = 1 + \tan^2(x)$ to express the remaining factors in terms of tangent. $u = \tan(x)$, $du = \sec^2(x) dx$
2. If m is odd, save a factor of $\sec(x) \tan(x)$ and use $\tan^2(x) = \sec^2(x) - 1$ to express the remaining factors in terms of secant. $u = \sec(x)$, $du = \sec(x) \tan(x) dx$
3. If n is odd and m is even, you likely need to use integration by parts.

Example: Find the antiderivative

$$\int \tan^3(x) \sec(x) dx$$

they look familiar

$$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

put $\tan^2$ in terms of $\sec$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\tan^2 \theta = \sec^2 \theta - 1$$

$$= u^2 - 1$$

$$dU = \sec(x) \tan(x) dx$$

$$U = \sec(x)$$

$$\int (u^2 - 1) du$$

$$= \frac{1}{3} u^3 - u + C = \frac{1}{3} \sec^3(x) - \sec(x) + C$$

Example: Find the antiderivative

$$\int \tan^2(x) \sec(x) dx$$

$$\tan^2(x) = \sec^2(x) - 1 \rightarrow \int (\sec^2(x) - 1) \cdot \sec(x) dx = \int \sec^3(x) dx - \int \sec(x) dx$$

$$= \int \sec^3(x) dx - \ln |\sec(x) + \tan(x)|$$

ln can't take a neg number hence why a abs val is needed

$$\int \sec^3(x) dx = \int \sec^2(x) \cdot \sec(x) dx$$

$$u = \sec(x) \quad du = \sec(x) \tan(x) dx$$

$$dv = \sec^2(x) dx \quad v = \tan(x)$$

$$\int \sec(x) \cdot \sec^2(x) dx = \sec(x) \tan(x) - \int \tan(x) \sec(x) \tan(x) dx = I$$

$$\sec(x) \tan(x) - I - \ln |\sec(x) + \tan(x)| = I$$

$$\sec(x) \tan(x) - \ln |\sec(x) + \tan(x)| = 2I$$

$$I = \frac{\sec(x) \tan(x) - \ln |\sec(x) + \tan(x)|}{2} + C$$

That's the OG question which we can now sub

Trigonometric Substitution: Whenever integrals have one of the following terms:

$$\sqrt{a^2 - x^2}, \quad \sqrt{x^2 - a^2}, \quad \sqrt{a^2 + x^2}$$

(where a is a positive number), we can use substitution with trig functions and Pythagorean theorem to simplify.

Example: If a is a positive number, determine

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$\sin \theta = \frac{x}{a} = \frac{x}{\sqrt{a^2 - x^2}}$
 $x \cos \theta = \frac{w}{a} = \frac{\sqrt{a^2 - x^2}}{a}$
 $\tan \theta = \frac{x}{w} = \frac{x}{\sqrt{a^2 - x^2}}$

$\int \sqrt{a^2 - x^2} dx$

$\rightarrow \int a \cos \theta dx$
 Need to put in terms of θ
 $x = a \sin \theta$
 $dx = a \cos \theta d\theta$

$\rightarrow \int \sqrt{a^2 - x^2} = \frac{w}{a} \cdot a = a \cos \theta$

$= a^2 \int \cos^2(\theta) d\theta = a^2 \int \frac{1}{2} (1 + \cos 2\theta) d\theta = \frac{a^2}{2} \cdot 1 - \frac{a^2}{2} \int \cos 2\theta d\theta$
 $u = 2\theta \quad du = 2d\theta \rightarrow d\theta = \frac{du}{2}$
 $\int \cos u \frac{1}{2} du = \frac{1}{2} \sin u = \frac{1}{2} \sin(2\theta)$
 $= \frac{a^2}{2} \left[\frac{1}{2} (-\sin(2\theta)) \right] + C = \frac{a^2}{2} \left[\theta + \frac{\sin 2\theta}{2} \right] + C$
 $= \frac{a^2}{2} \left[\arcsin\left(\frac{x}{a}\right) + \frac{1}{2} \sin\left(2 \arcsin\left(\frac{x}{a}\right)\right) \right] + C$

How are the sides determined?

Example: If a is a positive number, determine

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$\sin \theta = \frac{\sqrt{x^2 - 16}}{x}$
 $x \cos \theta = 4$
 $4 \tan \theta = \sqrt{x^2 - 16}$

How

$\int \frac{1}{\sqrt{x^2 - a^2}} dx$ say $a = 4$ so $\int \frac{1}{\sqrt{x^2 - 16}} dx$

$= \int \frac{1}{4 \tan \theta} dx$ $x = 4 \sec \theta$
 $dx = 4 \sec \theta \cdot \tan \theta d\theta$

$= \int \frac{1}{4 \tan \theta} 4 \sec \theta \tan \theta d\theta = \int \sec \theta d\theta$

$\theta = \arcsin\left(\frac{x}{4}\right)$ $\int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C$

How

$= \ln \left| \sec\left(\arcsin\left(\frac{x}{4}\right)\right) + \tan\left(\arcsin\left(\frac{x}{4}\right)\right) \right| + C$

How

$= \ln \left| \frac{x}{4} + \frac{\sqrt{x^2 - 16}}{4} \right| + C$

Can't go from 1 to the other, you'd pick one

$$\int \sqrt{x^2 + 16x} \, dx = \int \sqrt{x} \sqrt{x+16} \, dx$$

w/
+ rig
sub by u want
here to be
a square

$$u = \sqrt{x}$$

$$u^2 = x$$

$$du = \frac{1}{2u} dx$$

$$\Rightarrow \int 2u^2 \sqrt{u^2 + 16} \, du$$

and then you
can do + rig
sub